BSM|CS1T003/AI1T003/DS1T003|02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

Education to Eternity

B.Tech ~~Summer~~ / Winter - 2023

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: CSE/AI/DS **Sem:-I**

Subject:Basic Electrical and Electronics Engineering **Marks: 100**

Subject Code: CS1T003/AI1T003/DS1T003

Date : 13/01/2024 **Time: 3 Hrs.**

Instructions:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**
4. **Write proper S.I units for numerical.**

COs: Students should be able to:

CO1.Define fundamentals of electrical system and classify different wiring system.

CO2. Choose measuring instruments for measurement of electrical quantitiesand compare energy resources for electrical energy generation.

CO3. Understand the basic concepts of PN junction diode and its characteristics. Apply them in rectifiers and transducers.

CO4. Elaborate the transistor configuration in CE, CB & CC mode andapply transistor characteristics to construct amplifiers devices. Distinguish various logic gates and simplify the Boolean's equations.

CO5. Plan and organize the utilization of energy resources of electrical system &justify the utilization of various electrical and electronics components into electrical and electronics circuitries.

PARTA

Q. No.	Solve the two words or two sentences (All Questions are compulsory).	Marks	Bloom's Level and CO
Q.1	Define Electric Current.	2	L1/CO1
Q.2	List any two examples of positive temperature coefficient. conductor	2	L1/CO1
Q.3	Recall inductance with its S.I units.	2	L1/CO1
Q.4	State frequency of a signal.	2	L1/CO2
Q.5	Describe the secondary transmission.	2	L2/CO2
Q.6	Estimate the value of resistivity for copper?	2	L2/CO3
Q.7	Identify the type of rectifier which has highest efficiency. Also mention its percentage.	2	L2/CO3
Q.8	In a common base connection, I_E = 1mA, I_C = 0.95mA. Calculate the value of I_B.	2	L3/CO4
Q.9	List different types of basic logic gates.	2	L1/CO3
Q.10	Illustrate the advantages of the digital signal.	2	L2/CO3
Q.11	Identify the disadvantages of the A.C transmission.	2	L2/CO5
Q.12	Summarize the advantages of the D.C transmission.	2	L2/CO5

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PART B

Q.No.	Solve the following Descriptive Type Questions (All Questions are compulsory).	Marks	Bloom's Level and CO
Q.13	List the applications of variable type of inductors. (Any four)	4	L1/CO1
Q.14	Formulate the expression for Root Mean Square (R.M.S) value of the sinusoidal signal.	4	L6/CO2
Q.15	Evaluate the total work done by an electric energy source with a potential difference of 3kV that delivers a current of 1A for 1 minute?	4	L5/CO1
Q.16	In a common base connection, current amplification factor is 0.9. If the emittercurrent is 1mA, determine the value of base current.	4	L3/CO4
Q.17	Calculate the value of β if (i) α = 0.9 (ii) α = 0.98 (iii) α = 0.99.	4	L3/CO4
Q.18	In a common base connection, the emitter current is 1mA. If the emitter circuit isopen, the collector current is 50μA. Find the total collector current. Given that α = 0.92.	4	L3/CO4

PART C

Q. No.	Solve the following Descriptive Type Questions (Solve any 04 out of 06).	Marks	Bloom's Level and CO
Q.19	Demonstrate the casing and capping type of electrical wiring system with its advantages and disadvantages.	7	L3/CO1
Q.20	The temperature coefficient of resistance of a wire is 12.5×10^{-4} per degree Celsius at 300K. The resistance of the wire is 1Ω. The temperature at which the resistance is 2Ω is?	7	L4/CO1
Q.21	Classify sources of energy in detail with examples.	7	L4/CO2
Q.22	Explain Half Wave Rectifier (HWR) with the help of construction and working. Also mention its efficiency.	7	L4/CO3
Q.23	Survey different applications of Light Emitting diode and photo diode?	7	L4/CO3
Q.24	Discuss the construction and working of PNP transistor in detail?	7	L5/CO4

PART D

Q. No.	Solve the following Descriptive Type Questions (Solve any 02 out of 03).	Marks	Bloom's Level and CO
Q.25	Compare different types of Electrical wiring systems with the help of its parameters such as life, cost, mechanical protection, electrical protection, type of labour required, installation and popularity?	12	L5/CO1
Q.26	Modify a table comparing different transistor configurations (Common Base, Common Emitter and Common Collector) using its characteristics such as input resistance, output resistance, current gain, voltage gain and its applications?	12	L6/CO4
Q.27	Explain the structure of electric power system using single line diagram?	12	L5/CO5

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